

VISHAL RAJ

Software Development Engineer (Backend-leaning) | Node.js | REST APIs | DSA

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SUMMARY

Software Development Engineer with hands-on experience building backend-driven, production-ready applications using **Node.js**, **Express.js**, **REST APIs**, and databases such as **MongoDB** and **PostgreSQL**. Strong understanding of authentication, real-time systems (Socket.IO), API design, and data modeling. Proven problem-solving skills with **500+ DSA problems solved** across core algorithmic topics. Seeking **SDE and Backend** roles to build scalable systems and impactful products.

EDUCATION

National Institute of Technology, Andhra Pradesh

Bachelor of Technology in Electronics & Communication Engineering

2022 – 2026

CGPA: 8.54/10

EXPERIENCE

South Bihar Power Distribution Company Limited (SBPDCL) | Software Engineering Intern

May 2025 – June 2025

- Built a production-grade Car Marketplace platform supporting **100+ car listings**, advanced filtering, wishlisting, and end-to-end test drive booking workflows using **Supabase (PostgreSQL)**. **CERTIFICATE**
- Designed and integrated secure authentication and role-based access control with **Clerk**, enabling buyer, dealership admin, and internal team roles with booking state management.
- Integrated the **Google Gemini API** to extract structured car details from uploaded images, automating listing creation and reducing manual data entry. **LINK**

Indian Institute of Technology, Bhubaneswar | Research Intern (Remote)

Dec 2024 – Jan 2025

- Collaborated on a Microsoft-supported research internship to build a personalized travel planning system using structured geospatial and user-preference datasets, improving itinerary relevance and recommendation accuracy.
- Engineered rule-based backend logic to generate optimized **3, 5 and 7-day itineraries**. **CERTIFICATE**

PROJECTS

Realtime ChatApp | React, Socket.IO, Node.js, MongoDB

Mar 2025 – May 2025

- Architected a real-time messaging backend supporting **one-to-one and group chats** using **Socket.IO**, enabling low-latency bidirectional communication. **LINK**
- Secured REST and WebSocket endpoints with **JWT-based authentication**, including signup/login flows, password hashing, and session validation.
- Modeled **MongoDB schemas and APIs** for users, conversations, messages, group membership, and message persistence.

BookStore App | MERN Stack, Tailwind CSS

Jan 2025 – Feb 2025

- Built a backend-driven online bookstore managing **250+ books across 15+ genres**, exposing RESTful APIs for catalog browsing, search, and inventory management. **LINK**
- Enforced **secure JWT-based authentication** with encrypted password storage and **user-level authorization** to support free and paid book access.

SKILLS

Backend: Node.js, Express.js, REST APIs, Authentication (JWT), Socket.IO

Databases: MongoDB, PostgreSQL

Languages: C++, Python, SQL, Java (Basic)

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Developer Tools: Git, GitHub, Vercel

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, OOP

ACHIEVEMENTS / CERTIFICATIONS

• **Competitive Programmer** – LeetCode and GeeksforGeeks

- Solved **500+ Data Structures and Algorithms problems** across Graphs, Dynamic Programming, Trees, and Greedy Algorithms.

• **Top 10 Finalist** – Inspira 24-Hour Hackathon 2025

LINK

- Developed the **web application UI and Chrome extension** for a Fake News Detection system, integrating REST APIs exposed by a FastAPI backend serving a fine-tuned MobileBERT model.
- Implemented seamless **frontend-backend integration** for real-time predictions and integrated the **Google Gemini API** for factual content rewriting with CORS-enabled deployment.